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RESEARCH ARTICLE

State-of-Health Estimation for Lithium-Ion and Sodium-Ion Battery Cells Using Data-Driven Models and Cross-Chemistry Transfer Learning

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ABSTRACT The rapid expansion of green energy systems has accelerated the adoption of battery energy storage systems (BESS) and increased the demand for advanced battery management system (BMS). While lithium-ion batteries (LIBs) dominate current applications, concerns over resource limitations and cost have motivated growing interest in sodium-ion batteries (SIBs) as a more abundant and economical alternative. State of health (SOH) estimation remains essential for ensuring safe and reliable operation, yet most existing research focuses on LIBs due to the limited availability of high-quality SIB ageing datasets. This study conducted comprehensive ageing tests on LIBs and SIBs, obtained a unified dataset, and analyzed how their health declined over time, along with detailed patterns in their charging behavior, based on incremental capacity analysis (ICA) curve, to understand degradation. Using these features, seven deep-learning models, including deep neural networks (DNN), convolutional neural networks (CNN), long short-term memory (LSTM) networks, gated recurrent units (GRU), and three hybrid CNN–recurrent architectures were developed and evaluated under a Leave-One-Cell-Out (LOCO) scheme. For LIBs, recurrent and CNN–recurrent hybrid models achieved the best accuracy, reaching a minimum mean absolute error (MAE) of 0.8811%. In contrast, simpler architectures such as DNN and CNN were more effective for SIBs, with the best model achieving a minimum MAE of 0.2591%. Transfer learning (TL) from LIB to SIB demonstrated promising feasibility: several model–cell combinations showed unchanged or improved accuracy, although larger degradation-pattern mismatches in some SIB cases led to MAE increases of up to 1.4106%. Overall, the findings highlight shared electrochemical ageing characteristics between LIBs and SIBs and demonstrate the potential of leveraging extensive LIB datasets to support SOH estimation for emerging SIB technologies.

INDEX TERMS Sodium battery cell, data-driven, state of health, incremental capacity analysis, transfer learning.

I. INTRODUCTION

As the global share of sustainable energy gradually increases, battery energy storage systems (BESS) are becoming increasingly important. BESS systems have been successfully applied in various sectors, including automotive, industrial, and power grids [1]. In the existing battery chemistry

system, lithium-ion batteries (LIBs), especially lithium iron phosphate (LFP) batteries, dominate the market due to their high energy density, and excellent cycle life [2]. However, the rapid growth in lithium demand, resource scarcity, and cost volatility have raised concerns about long-term sustainability [3]. To address these challenges, sodium-ion batteries (SIBs) have emerged as a promising alternative, drawing increasing academic and industrial interest due to the natural abundance and low cost of sodium resources, which

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make them particularly appealing for large-scale energy storage and other cost-sensitive applications [4], [5], [6]. However, SIBs still face some limitations, such as lower energy density, more significant structural changes during cycling, and a less mature commercial application compared to LIBs [7], [8], [9].

To fully realize the performance of a BESS, a powerful and intelligent battery management system (BMS) is essential [10]. An efficient BMS maintains safe and efficient operation by continuously assessing battery state, regulating charge balance, identifying anomalies, managing thermal dynamics, and preventing operations outside safe limits [11], [12]. These key functions rely on accurate state estimation, with state of charge (SOC), state of health (SOH), and remaining useful life (RUL) [13], [14], [15], [16] being a crucial indicator of battery aging and long-term reliability. In this work, the SOH is adopted as the primary indicator of capacity degradation and is mathematically defined as:

$$\text{SOH} = \frac{Q_t}{Q_0} \times 100\%. \quad (1)$$

Here, Q_t denotes the current maximum capacity at time t , while Q_0 represents the initial maximum capacity of the battery at the beginning-of-life.

Battery SOH estimation methods can be broadly classified into experimental, model-based, and data-driven approaches. Experimental methods infer SOH from measurable quantities such as internal resistance, impedance, or full-charge capacity, but they typically require complete charge–discharge cycles or intrusive tests, limiting their real-time applicability [17], [18]. Model-based methods estimate SOH by identifying ageing-sensitive parameters within electrochemical or equivalent-circuit models [19]. Although model-based health assessments rely on physically reliable battery models and have highly accurate electrochemical formulas, their structural complexity [20] computational cost make them difficult to apply in practical applications in resource-constrained BMS [21]. With the rapid advancement of machine learning (ML) in different fields, data-driven approaches have attracted increasing research interest in battery SOH estimation. As they do not require in-depth electrochemical descriptions of the battery aging process and therefore reduce the modeling burden, data-driven methods have become increasingly popular; however, their predictive performance remains constrained by the richness and representativeness of the training data and is often sensitive to model-parameter settings [22], [23], [24], [25].

Among them, incremental capacity analysis (ICA) with deep learning (DL) models has emerged as a promising technique for SOH estimation. This method extracts ICA as input features for DL models to estimate SOH. The characteristic peaks and valleys of the ICA curve have been demonstrated to correlate strongly with capacity fade [26], [27]. ICA-based data-driven models have become widely adopted for battery SOH estimation. Sun et al. [28] proposed a hybrid

approach that integrates ICA-derived health indicators with a recurrent neural network (RNN) for LIB SOH estimation, while Wang et al. [29] further improved ICA-based prediction accuracy by accounting for the effects of varying charge–discharge rates. Recently, Eleftheriadis et al. [30] compared various ML methods across public datasets using ICA features derived from partial discharges. Chen et al. [31] further explored ICA-based feature representations in a multidimensional space, applying Bayesian optimization to identify informative features for SOH estimation.

Although data-driven DL approaches have demonstrated strong performance in battery health estimation, most existing studies focus primarily on lithium-ion batteries, while research on sodium-ion batteries remains relatively limited [32]. One reason is that SIB technology is still in its early stages of development, its long-term aging datasets and performance degradation mechanisms are not as sophisticated as those of LIB. Most existing ICA studies focus solely on LIB, given the fact that SIB data are limited while extensive LIB datasets are available. For example, the most widely used public datasets, such as the NASA UBD [33], CALCE [34], Oxford Battery Degradation Dataset [35], are all based exclusively on LIB. Furthermore, commercial sodium-ion batteries employing hard carbon anodes exhibit more uniform ICA curves and more linear voltage profiles over a wider voltage range, primarily due to the absence of well-defined phase transitions in hard carbon, in contrast to the pronounced phase-transition behavior observed in graphite-based lithium-ion batteries [36]. This characteristic leads to less distinctive peaks in the ICA curves compared with LIBs, making peak-based feature extraction more challenging [37].

To bridge this research gap, transfer learning provides a promising approach by reducing the reliance on large SIB datasets under data-scarce conditions, while leveraging the rich degradation information learned from LIBs. This enables the model to extract transferable ICA features and ultimately improves SOH estimation performance.

Transfer learning leverages a previously trained model on the source task as a foundation to learn new, relevant target tasks. By reusing the knowledge encoded in the pre-trained model, this approach can significantly reduce training workload and computational cost. This method is particularly advantageous when the target dataset is limited, as the transferred features help alleviate data scarcity and improve learning efficiency [38], [39], [40].

However, research on the transferability of ICA models between LIB and SIB systems is limited. Transfer learning offers a compelling solution: by reusing knowledge learned from data-rich LIB systems, it reduces reliance on large SIB datasets, shortens training time, and enhances model robustness in situations where data scarcity or distribution variations hinder traditional data-driven approaches for SIB SOH estimation. Overall, the main contributions of this study are:

- Apply transfer learning to translate knowledge from LIB to SIB, enabling the assessment of model robustness and cross-chemistry generalization.
- Investigate the impact of ICA features for accurate SOH estimation of SIB using data-driven modeling techniques.
- Develop and evaluate multiple DL architectures, benchmarking their performance across both LIB and SIB datasets under diverse operating conditions.

The remainder of this paper is organized as follows. Section II explains what dataset we used through the experiments and proposed ICA processing pipeline. Section III presents the modeling framework, and the transfer learning strategy from LIB to SIB. Section IV reports the experimental findings, including comparative analyses between LIB and SIB, padding effects, and the performance of cross-chemistry transfer learning. Finally, Section V summarizes the key outcomes and discusses future research directions for ICA-based SOH estimation.

II. DATASET

A. DATA COLLECTION

The dataset was collected through controlled laboratory experiments conducted on three SIBs (3 Ah, 3.0 V, 26700 form Smart Electronics) and four Li-ion, LFP (1.8 Ah, 3.2 V, 18650 form NX). Representative photographs of the tested Li-ion and Na-ion cells are shown in Fig. 1.

All experiments were performed using a programmable battery testing system in conjunction with a thermal chamber to ensure stable and repeatable operating conditions. The experiments were configured and manually controlled via a dedicated PC-based application, which allows flexible programming of test protocols and real-time monitoring. The experimental setup is illustrated in Fig. 1, while the main specifications of the testing equipment are summarized in Table 1.

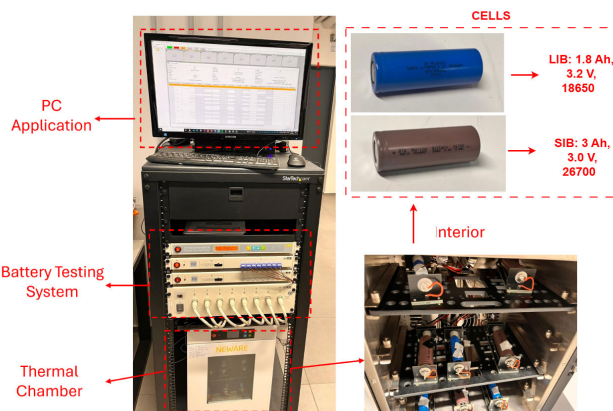


FIGURE 1. Equipment setup.

The collected database includes 23 HPPC tests, as well as 23 ageing procedures for both LIBs and SIBs. Each profile

TABLE 1. Experimental equipment used for battery testing and thermal control.

Parameter	Battery Thermal Chamber	Battery Testing System
Model	MHW-25	CT-4008T-5V12A-204n
Temperature range	15–60 °C	–
AC input	220 V, 50/60 Hz	220 V, 50/60 Hz
Max. power	0.2 kW	850 W

contains voltage, current, capacity, temperature, and time data, recorded with a sampling interval of 1 s.

Every 30 cycles, a capacity characterization test was performed in order to capture the current maximum capacity of each battery cell, information that is necessary for providing cells' true SOH value. As shown in Fig. 2, the LIBs exhibit a relatively linear and faster degradation trend, dropping from approximately 100% to below 90% for most cells. Particularly, Cell 8 declines to below 80%. In contrast, the SIBs display a more complex SOH evolution with noticeable fluctuations with capacity regeneration, which may be attributed to the more intricate electrochemical and mechanical behaviors of Na-ion chemistry under different driving modes. Similar degradation behavior for SIBs can be seen in [41]. Nevertheless, the overall degradation rate of SIBs is slower. Only Cell 6 falls slightly below 90% after 23 ageing cycles, while Cell 7 stabilizes around 92% and Cell 2 retains approximately 94% SOH at the end of the campaign. These results clearly indicate that SIBs demonstrate more resilient SOH degradation compared with LIBs.

The dataset used in this study comes from a comprehensive battery degradation campaign designed to characterize the static and dynamic performance of LIBs and SIBs. As shown in Fig. 3, the campaign comprises three main parts: characterization testing, hybrid pulse power characteristic (HPPC) measurements, and long-term aging under real-world driving profiles. These parts together can provide the true SOH values and help with ICA curve formation.

1) CAPACITY CHARACTERIZATION TEST (1C)

Each degradation cycle begins with a constant-current-constant-voltage charging profile, which is named as CC–CV step, carried out in accordance with the manufacturer's recommended protocol to ensure complete and optimal charge. During the CC phase, the battery is charged at a fixed 1C current until the upper voltage threshold is reached, after which the CV phase maintains this voltage until the current reaches its cut-off value. A subsequent 1C discharge (CC) is performed to measure the current maximum capacity, followed by a 15-minute rest and a second CC–CV charge to restore the cell to a well-defined fully charged state. Finally, the cell rests for one hour to reach thermal and electrochemical equilibrium before entering the dynamic testing stage.

2) HYBRID PULSE POWER CHARACTERISTIC TEST

The HPPC test is then conducted to measure dynamic internal resistance, polarization behaviour, and open-circuit

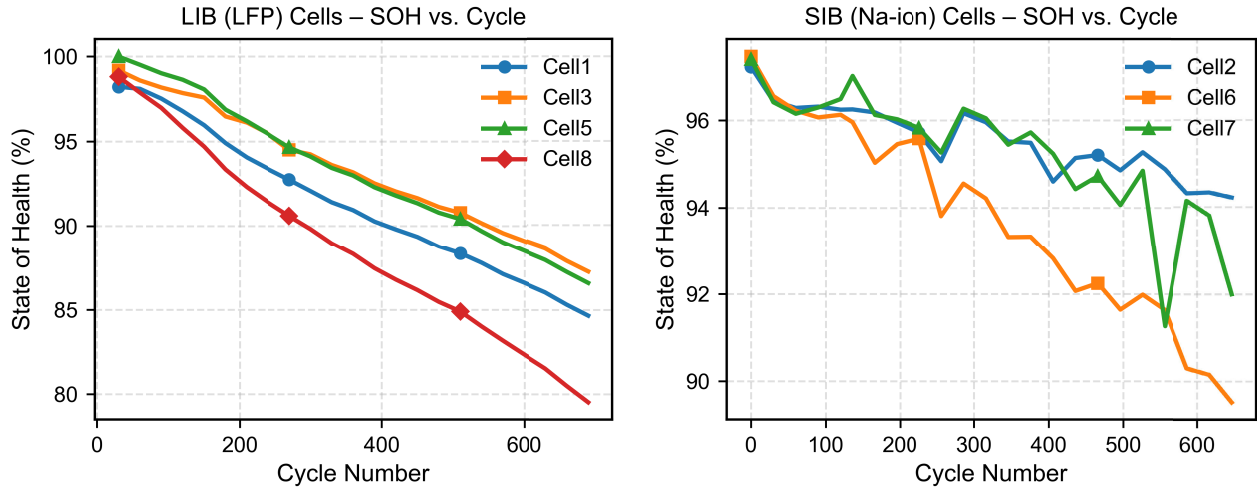


FIGURE 2. Comparison of SOH degradation trends for (a) LFP cells and (b) Na-ion cells.

voltage (OCV) characteristics [42]. The process involves repeatedly starting with a 1-hour reset, followed by a short CC discharge/charge pulse, then a 10% SOC CC discharge. This pattern is repeated until ten cycles have been completed and the SOC is close to 0%.

3) DYNAMIC PROFILES

Following HPPC testing, the cells are subjected to dynamic driving profiles (e.g., LA92, US06, UDDs, Mixed 1–7). These load patterns simulate the high-frequency current fluctuations experienced by electric vehicles during operation and impose realistic dynamic stress on the battery. Under more aggressive driving conditions, the battery is subjected to greater load changes and therefore tends to degrade more quickly [43]. After each 30-cycle driving cycle ageing simulation is completed, a new characteristic test is performed, followed by a new HPPC test to update the battery's health data, and then the next ageing procedure is started again.

Collectively, the characterization tests, HPPC tests, and driving-cycle ageing workflow generate a rich degradation dataset that captures the comprehensive characteristics of both LIBs and SIBs, thereby enabling deeper analyses such as SOH estimation.

B. ICA FEATURE EXTRACTION

In this study, ICA features were extracted from the constant-current (CC) charge segments of the HPPC test files, using [44] as a reference. Under CC conditions, the whole life cycle of battery pack is relatively stable to extract the features [45]. Also, the current is fixed during this stage, the ICA curve can be reliably computed by numerically differentiating the measured Q - V relationship [46]. The ICA values were obtained through a finite-difference approximation of the form

$$\left. \frac{dQ}{dV} \right|_i \approx \frac{Q_{i+1} - Q_{i-1}}{V_{i+1} - V_{i-1}},$$

applied to the smoothed $Q(V)$ data. This method can generate stable ICA curves, capture the evolution of electrochemical behavior during aging cycles, and form a core dataset for feature extraction, model training, and cross-chemical transfer learning.

As illustrated in Fig. 4, during the CC charging phase, both LIB and SIB are operated at constant currents (0.5 C-rate). However, the voltage rise patterns of the two batteries are significantly different, indicating that the resulting ICA curves should also exhibit different structures. Specifically, SIBs exhibit a wider voltage range compared with LIBs, and LIBs show a sloping plateau region, while SIBs present a more linear voltage profile, which corresponds to the differences in their underlying electrochemical mechanisms discussed in Section I.

In order to minimize the numerical noise introduced by the derivative (which amplifies high-frequency measurement noise), signal smoothing is essential for providing a better ICA curve for subsequent SOH estimation [47]. Various filtering techniques are widely used in the literature, including Gaussian smoothing [48], [49], and Savitzky-Golay (SG) filtering [50], [51]. In this work, the ICA smoothing was performed with SG filter. For a given window length L and polynomial order p , the SG operation is defined as

$$\hat{y}_i = \sum_{j=-m}^m c_j y_{i+j}, \quad m = \frac{L-1}{2},$$

where $\hat{y}_i$ is the smoothed output and c_j are the convolution coefficients computed from a local polynomial regression. Specifically, within each moving window, the sample values y_{i+j} (for $j = -m, \dots, m$) are approximated by a polynomial of degree p ,

$$y(i+j) \approx \sum_{k=0}^p b_k j^k,$$

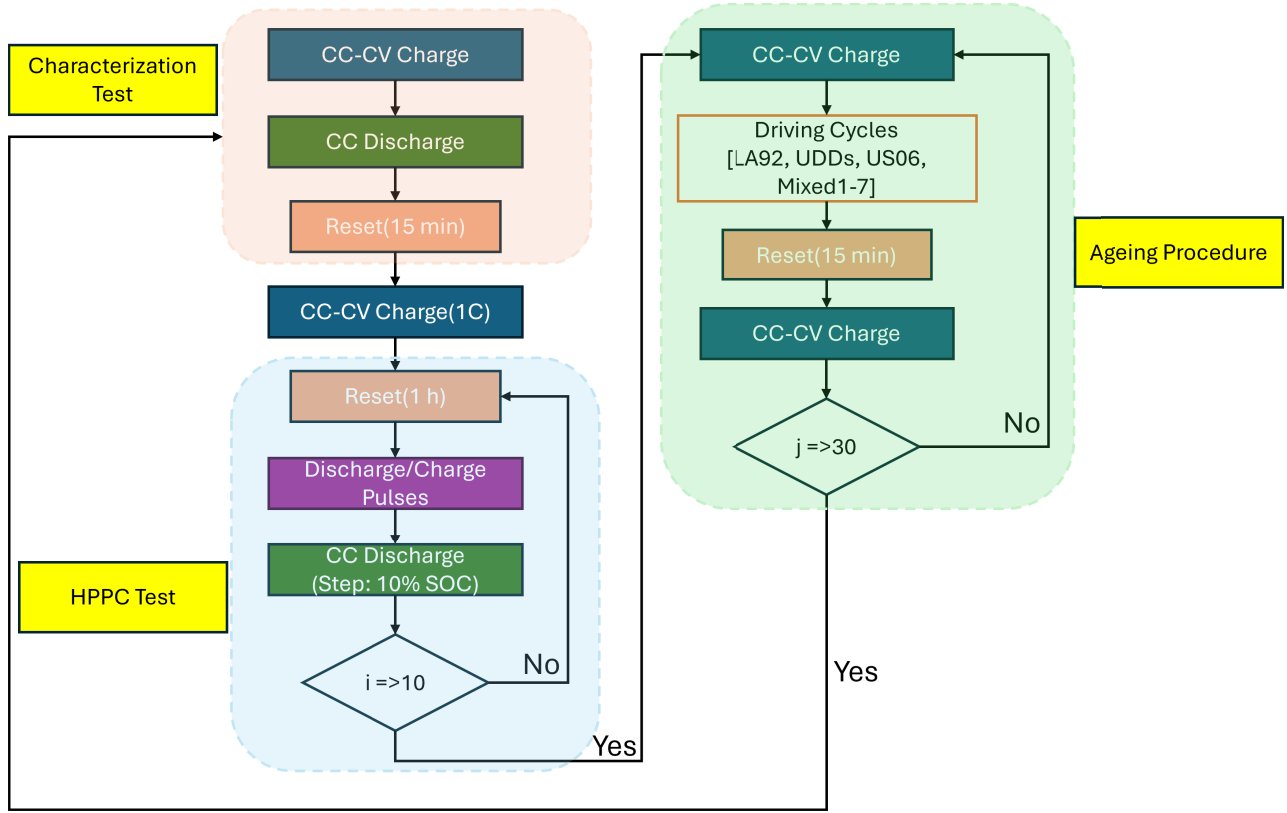


FIGURE 3. Workflow of battery degradation campaign.

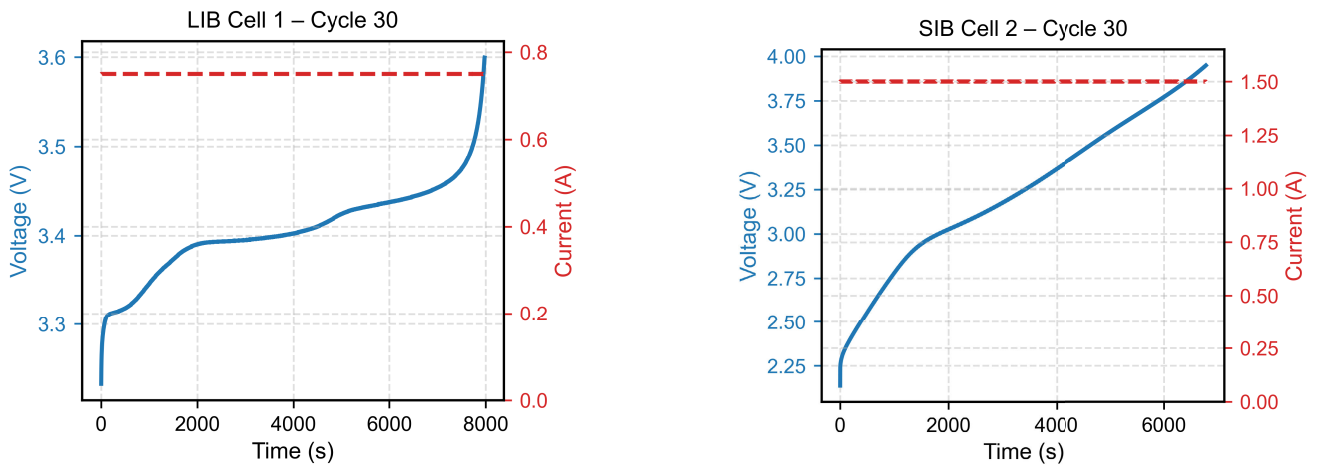


FIGURE 4. Comparison of the constant-current phase during the CCCV charge for (a) LIB Cell 1 and (b) SIB Cell 2 at Cycle 30.

and the coefficients b_k are obtained by least-squares fitting. In this formulation, the fitted value at the center of the window corresponds to the constant term of the polynomial,

$$Y(0) = b_0,$$

which represents the local estimate of the signal at position i . This fitted center value b_0 is exactly equal to the SG output $\hat{y}_i$, i.e., $\hat{y}_i = b_0$. The resulting SG coefficients c_j therefore

define the linear mapping from the raw samples to this center-point estimate b_0 , thereby enabling simultaneous smoothing and preservation of local peak features.

In this work, the SG filter was applied in two stages, consisting of a pre-smoothing step followed by a post-smoothing step. The pre-filter was used to further reduce measurement noise prior to numerical differentiation [52].

Previous studies on battery SOH estimation have adopted various SG configurations. For example, a window length of 9 with a polynomial order of 3 has been used in several ICA-based studies [47], while a larger configuration with window length of 21 and polynomial order of 3 has also been reported [53]. In addition, the SG window length is typically chosen as an odd number to avoid boundary artifacts during filtering [54].

In our study, the LIB datasets exhibited relatively clean $Q(V)$ and dQ/dV profiles. Therefore, a standard SG configuration was adopted (SavGol), using a short pre-window ($L_{\text{pre}} = 7$, $p_{\text{pre}} = 2$) and a moderate post-window ($L_{\text{post}} = 15$, $p_{\text{post}} = 3$).

In contrast, SIBs showed stronger voltage ripple and capacity fluctuation during charge. To stabilize the ICA curves under these noisier conditions, a stronger smoothing configuration (SavGol-Larger) was employed, with an extended pre-window ($L_{\text{pre}} = 25$) and a long post-window ($L_{\text{post}} = 55$), both using a third-order polynomial ($p = 3$).

Comparisons between the raw, standard-SG, and large-SG ICA curves (Fig. 5 and Fig. 6) show that applying an oversized SG window to LIB data leads to excessive smoothing, resulting in an unnatural reduction of peak amplitude and loss of fine electrochemical features. In contrast, the SIB ICA curves benefit substantially from the larger SG window: the increased smoothing produces a more stable and interpretable profile while preserving the essential peak structure despite the higher intrinsic noise level. Based on these observations, the SavGol was selected for the LIB dataset, whereas the SavGol-Larger was adopted for the SIB dataset in all subsequent feature extraction and SOH estimation procedures.

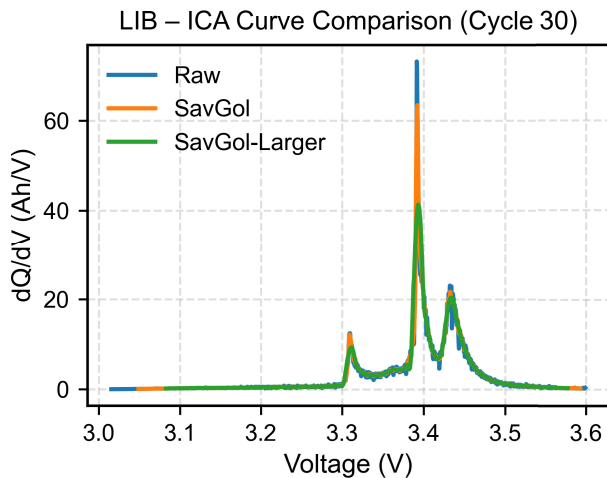


FIGURE 5. Comparison of LIB ICA curves under different SG configurations.

Finally, we analyzed the complete ICA evolution process of LIB and SIB, as shown in Fig. 7 and Fig. 8. Representative cells LIB 1 and SIB 2 were selected to illustrate the evolution of characteristic electrochemical peaks during

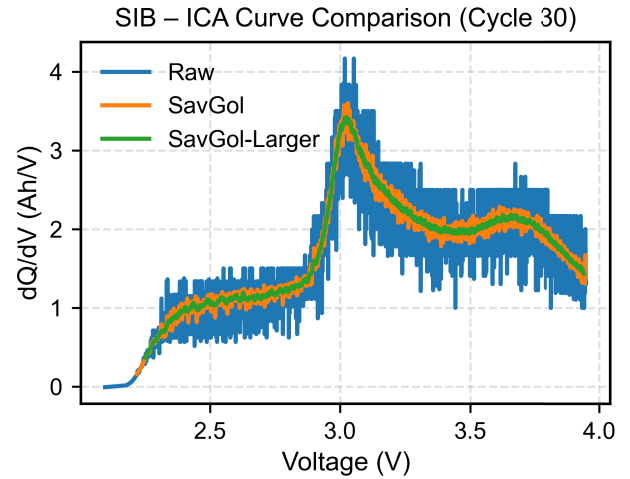


FIGURE 6. Comparison of SIB curves under different SG configurations.

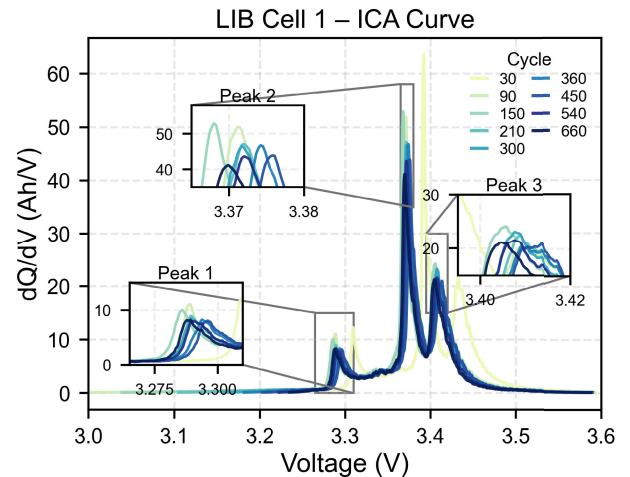


FIGURE 7. ICA evolution of the LIB across selected cycles with zoomed views of the characteristic ICA peaks.

specific cycling processes. Although the ICA curves of selected LIB and SIB exhibit different characteristics, both systems show a gradual evolution of the ICA curve during cycling, especially in terms of the position and value of the main peak. Specifically, the peak shape of LIB is sharper and clearer, with two additional sub-peaks, reflecting the more complex phase transition behavior of the electrode material. In contrast, the ICA curve of SIB is relatively broad, with only one main peak, indicating that its electrochemical reaction characteristics are different. These observations are consistent with the electrochemical characteristics discussed in Section I.

Specifically, the dominant ICA peak of the LIB exceeds 60 Ah/V at Cycle 30. As cycling progresses, the ICA profile gradually shifts, and by Cycle 660 the peak decreases to below 45 Ah/V, indicating changes in the underlying

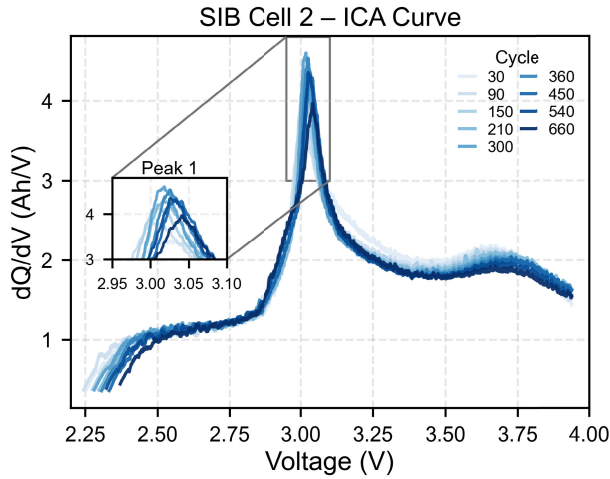


FIGURE 8. ICA evolution of the SIB across selected cycles with zoomed views of the characteristic ICA peaks.

electrode reaction mechanisms and a reduction in the amount of available active material [55].

A similar degradation trend is observed for the SIB, although the evolution pattern differs from that of the LIB. In the early cycles, the SIB exhibits ICA peaks below 4 Ah/V. As cycling proceeds, the peak value first increases, reaching values above 4.5 Ah/V, before gradually decreasing again toward Cycle 660. Moreover, the overall ICA profiles also show a systematic shift as the cycle number increases.

While ICA captures the detailed evolution of electrochemical processes through peaks and valleys in the voltage-capacity response, simpler scalar indicators such as internal resistance provide only limited information about cell degradation. In particular, resistance evolution tends to be non-monotonic and does not directly reflect the underlying reaction mechanisms or phase transitions. This is clearly observed in Fig. 9, where resistance trajectories show non-monotonic and cell-specific fluctuations for both LIB and SIB, reflecting the influence of multiple overlapping phenomena such as polarization, temperature effects, and contact resistances rather than distinct electrochemical reactions. Consequently, resistance fails to provide a consistent representation of ageing across chemistries, limiting its usefulness for transfer learning. In contrast, ICA features retain structured signatures linked to phase transitions and reaction mechanisms, which are more comparable between LIB and SIB systems and therefore better suited for learning transferable degradation patterns.

III. METHODOLOGY

In the section, all models were trained under a unified data-processing pipeline and a Leave-One-Cell-Out (LOCO) evaluation framework for both LIBs and SIBs. In each run, one cell was held out for testing, while the remaining cells were randomly split into 80% for training and 20% for

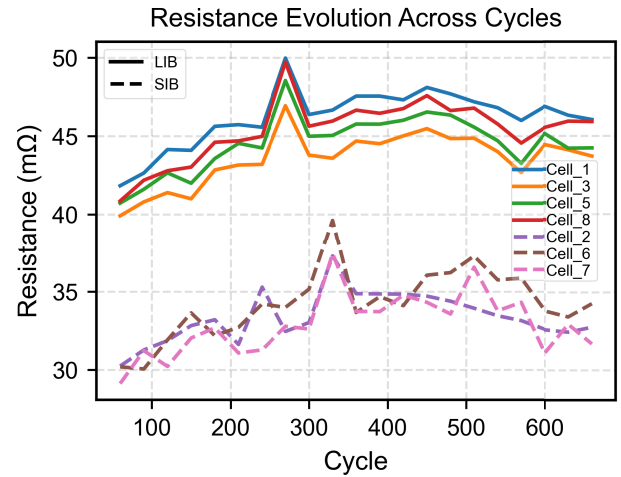


FIGURE 9. Resistance evolution across cycles for LIB and SIB cells. The resistance exhibits irregular behavior across cells, highlighting its limited ability to capture detailed electrochemical degradation patterns.

validation. This strategy enables a rigorous assessment of the models' generalization capability across different cells.

A. INPUT SHAPE FORMATION

Different cycles and different cells exhibit varying voltage ranges, therefore the resulting ICA sequences have different lengths. For this reason, sequence standardisation is required before model training. Common approaches include padding with masking or direct truncation. In this work, a truncation strategy was adopted, since the truncated ICA curves still preserve the dominant feature patterns (i.e., the peaks), and preliminary experiments showed that truncation outperformed the padding–masking approach.

For each battery chemistry, all ICA curves were truncated to the global minimum sequence length observed within that chemistry, producing fixed-length ICA feature vectors ($L_{\text{LIB}} = 849$ and $L_{\text{SIB}} = 3953$). This preprocessing ensures consistent model inputs and enables fair cross-cell comparisons.

To ensure numerical stability, the ICA inputs were standardized using Z-score normalization:

$$X_{\text{norm}} = \frac{X - \mu_{\text{train}}}{\sigma_{\text{train}}},$$

where μ_{train} and σ_{train} denote the mean and standard deviation computed solely from the training data to prevent information leakage. The same scaling parameters were then applied to the test cell.

B. MODELS' ARCHITECTURE CONFIGURATION

Seven neural network–based deep learning architectures commonly employed for battery SOH estimation were considered, including feedforward models such as deep neural networks (DNNs) and convolutional neural networks (CNNs), which are particularly effective at extracting com-

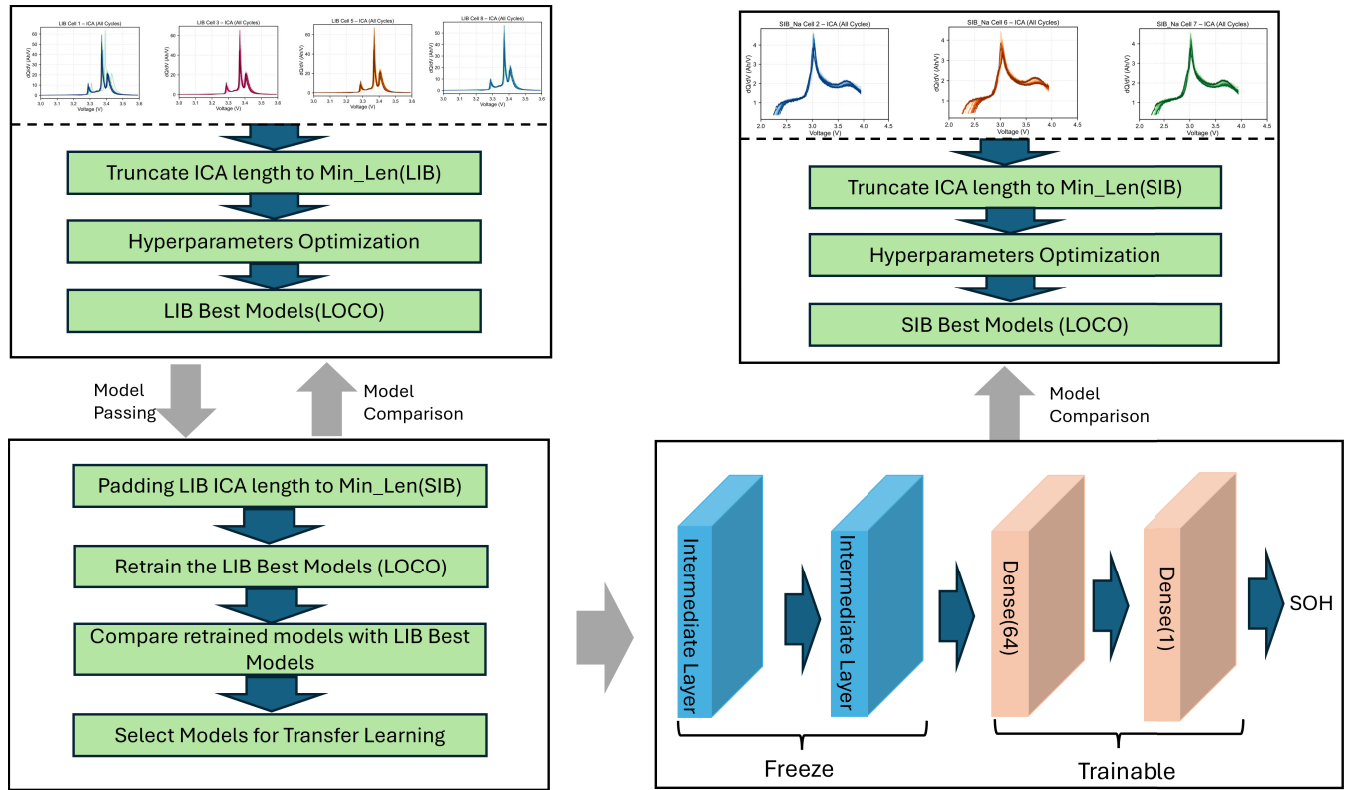


FIGURE 10. Overall workflow for ICA truncation, individual model optimization, padding/masking-based re-training, and cross-chemistry transfer learning.

plex features, as well as recurrent architectures such as long short-term memory (LSTM) networks and gated recurrent units (GRUs), which are well suited for capturing temporal dependencies in sequential data [56]. In addition, hybrid models, including CNN-LSTM, CNN-GRU, and CNN-BiLSTM, were investigated to leverage the complementary strengths of different architectures [57]. These models were selected to enable a comprehensive comparison between architectures specialized in feature extraction and those designed to capture temporal dependencies.

Hyperparameter selection was conducted separately for LIB and SIB under an LOCO framework. The hyperparameter search included two learning rates (10^{-3} and 5×10^{-4}), a fixed batch size of 2, and network widths drawn from the hidden-dimension set 256, 128, 64, 32. Structural variations were applied only to the feature-extraction layers. For DNN models, all two-layer combinations of these hidden sizes were considered, along with a three-layer configuration ([256, 128, 64]). For CNN architectures, the same size pairs were used as convolutional filter widths. For LSTM and GRU networks, all two-layer combinations of recurrent units selected from the same size set were evaluated. Hybrid models were formed by combining one of four convolutional filter pairs, [64, 128], [128, 64], [128, 128], or [256, 128], with a recurrent layer containing either 64 or 128 units.

All architectures share a common output layer consisting of two fully connected layers (64 units and one 1-unit layer). Each candidate configuration is trained using the Adam optimizer for a maximum of 200 epochs, with an early stopping strategy (patience of 30) applied to minimize the mean squared error (MSE). Model selection is performed based on the mean absolute error (MAE) computed on the validation set, and the configuration achieving the lowest MAE is retained as the optimal model for the corresponding test unit. The coefficient of determination (R^2) is additionally reported as a complementary performance indicator to assess the goodness of fit between the predicted and measured values.

The MSE is defined as

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2, \quad (2)$$

where y_i and $\hat{y}_i$ denote the ground-truth and predicted values of the i -th sample, respectively, and N is the total number of samples.

The MAE is defined as

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|. \quad (3)$$

TABLE 2. Performance Summary of All Models on LIBs (1, 3, 5, 8). All MAE values represent the real, unnormalized MAE in SOH[%].

Test Cell	Model	Val MAE	Test R ²	Test MAE	Best Structure (with lr)
1	DNN	0.5502	0.5263	1.8741	layers:[128,32], lr=0.001
	CNN	0.6085	0.5184	1.8288	filters:[256,64], lr=0.0005
	LSTM	0.9890	0.8392	1.3267	units:[256,64], lr=0.001
	GRU	1.1018	0.6675	2.0194	units:[128,32], lr=0.0005
	CNN_LSTM	0.4999	0.6622	1.7187	cnn:[128,64], lstm:128, lr=0.0005
	CNN_GRU	0.7868	0.6472	1.9479	cnn:[64,128], gru:128, lr=0.0005
	CNN_BiLSTM	0.4355	0.6685	1.7507	cnn:[128,64], bilstm:128, lr=0.0005
3	DNN	0.7838	0.8030	1.3687	layers:[256,128], lr=0.001
	CNN	1.1786	0.8646	1.1457	filters:[256,32], lr=0.001
	LSTM	0.7443	0.9069	0.9367	units:[256,32], lr=0.001
	GRU	1.5808	0.9162	0.9128	units:[256,32], lr=0.0005
	CNN_LSTM	1.2780	0.8871	1.0259	cnn:[64,128], lstm:64, lr=0.001
	CNN_GRU	1.2489	0.8880	1.0447	cnn:[64,128], gru:128, lr=0.0005
	CNN_BiLSTM	0.7137	0.8997	1.0248	cnn:[128,128], bilstm:64, lr=0.0005
5	DNN	1.2161	0.9039	1.0472	layers:[256,64], lr=0.0005
	CNN	1.4539	0.8658	1.1930	filters:[256,64], lr=0.001
	LSTM	1.4633	0.8758	1.2299	units:[256,128], lr=0.0005
	GRU	1.7315	0.8285	1.5693	units:[64,32], lr=0.0005
	CNN_LSTM	1.4268	0.8881	1.2035	cnn:[256,128], lstm:128, lr=0.001
	CNN_GRU	1.5972	0.8080	1.5695	cnn:[128,64], gru:128, lr=0.0005
	CNN_BiLSTM	1.3147	0.9413	0.8811	cnn:[128,128], bilstm:128, lr=0.001
8	DNN	0.7447	0.7064	2.3438	layers:[256,128,64], lr=0.001
	CNN	0.6800	0.7414	2.0931	filters:[256,128], lr=0.001
	LSTM	0.7825	0.7177	2.1903	units:[256,32], lr=0.001
	GRU	0.7704	0.7182	2.1500	units:[128,32], lr=0.0005
	CNN_LSTM	0.7140	0.7479	2.0497	cnn:[256,128], lstm:128, lr=0.001
	CNN_GRU	0.7886	0.7320	2.0696	cnn:[128,64], gru:64, lr=0.001
	CNN_BiLSTM	0.7485	0.7180	2.1760	cnn:[128,64], bilstm:64, lr=0.001

TABLE 3. Performance Summary of All Models on SIBs (2, 6, 7). All MAE values represent the real, unnormalized MAE in SOH[%].

Test Cell	Model	Val MAE	Test R ²	Test MAE	Best Structure (with lr)
2	DNN	0.1949	0.7126	0.3203	layers:[128,64], lr=0.0005
	CNN	0.1920	0.8041	0.2591	filters:[64,32], lr=0.001
	LSTM	0.3218	0.6017	0.3539	units:[256,32], lr=0.0005
	GRU	0.3109	0.6071	0.3684	units:[128,32], lr=0.0005
	CNN_LSTM	0.2886	0.6632	0.3417	cnn:[128,128], lstm:64, lr=0.0005
	CNN_GRU	0.3099	0.4770	0.4076	cnn:[256,128], gru:128, lr=0.001
	CNN_BiLSTM	0.3173	0.6660	0.3203	cnn:[256,128], bilstm:64, lr=0.0005
6	DNN	0.1070	0.9065	0.4281	layers:[64,32], lr=0.0005
	CNN	0.1053	0.8514	0.6054	filters:[128,64], lr=0.001
	LSTM	0.1701	0.2757	1.4140	units:[128,64], lr=0.0005
	GRU	0.1916	0.4260	1.2328	units:[256,128], lr=0.001
	CNN_LSTM	0.1607	0.6164	1.0293	cnn:[128,64], lstm:128, lr=0.0005
	CNN_GRU	0.1739	0.4537	1.2477	cnn:[128,128], gru:128, lr=0.0005
	CNN_BiLSTM	0.1487	0.4235	1.2766	cnn:[128,128], bilstm:64, lr=0.001
7	DNN	0.1345	0.7543	0.3888	layers:[256,32], lr=0.001
	CNN	0.1843	0.7646	0.3487	filters:[256,128], lr=0.001
	LSTM	0.3337	0.6251	0.5879	units:[128,32], lr=0.0005
	GRU	0.3331	0.6439	0.5785	units:[128,32], lr=0.0005
	CNN_LSTM	0.3473	0.5643	0.6409	cnn:[128,128], lstm:64, lr=0.0005
	CNN_GRU	0.3348	0.6376	0.5985	cnn:[256,128], gru:128, lr=0.0005
	CNN_BiLSTM	0.3112	0.5954	0.6295	cnn:[128,64], bilstm:128, lr=0.0005

The coefficient of determination (R^2) is defined as

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}, \quad (4)$$

where $\bar{y}$ denotes the mean of the ground-truth values.

C. TRANSFER LEARNING (LIB TO SIB)

Among various deep transfer learning strategies, fully fine-tuned and feature-based layer-freezing transfer remain the most widely adopted methods because they are simple, stable, and effective in scenarios with limited data [40].

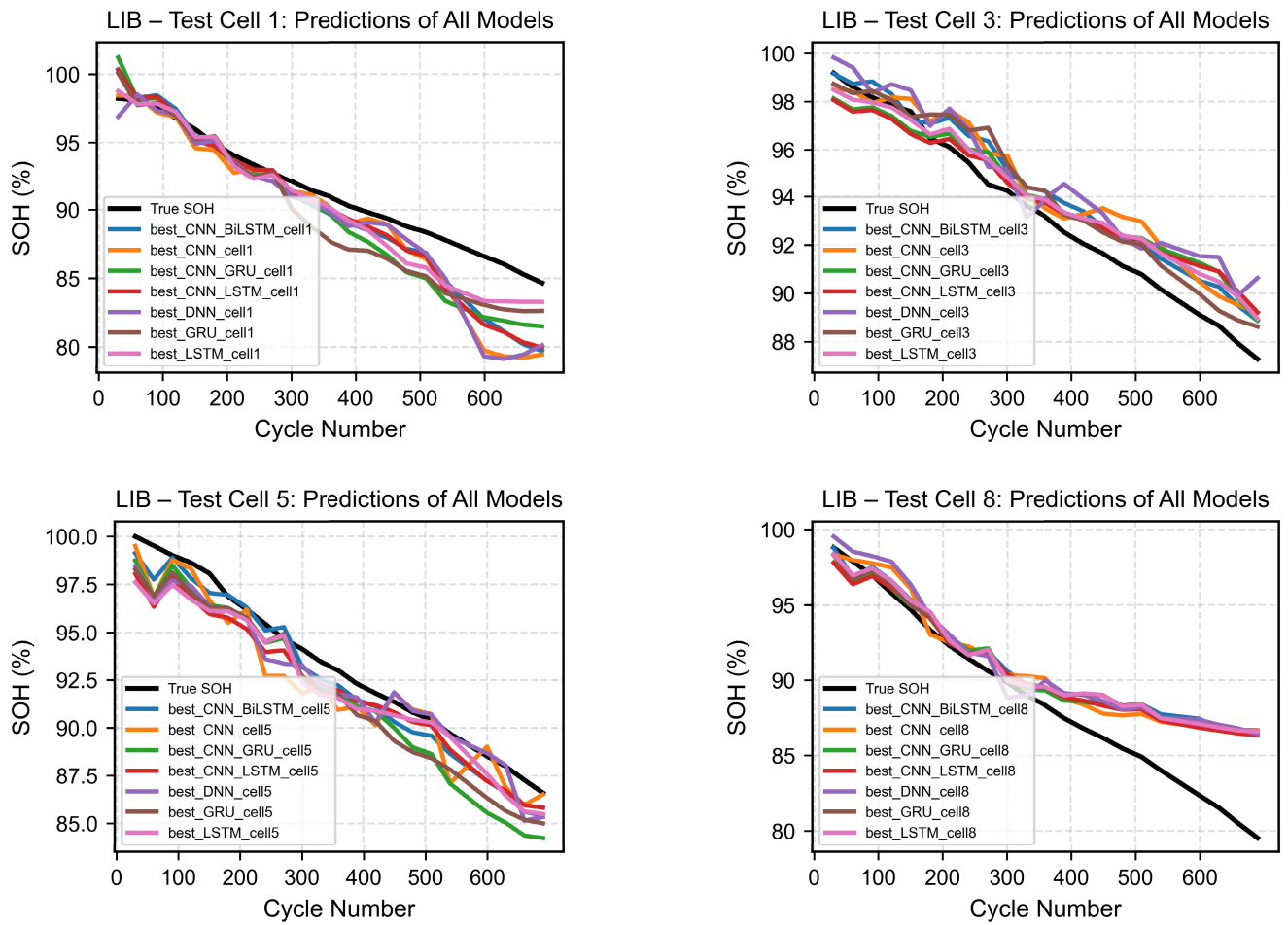


FIGURE 11. Predictions of all models for the four LIB test cells: Cell 1 (top-left), Cell 3 (top-right), Cell 5 (bottom-left), and Cell 8 (bottom-right).

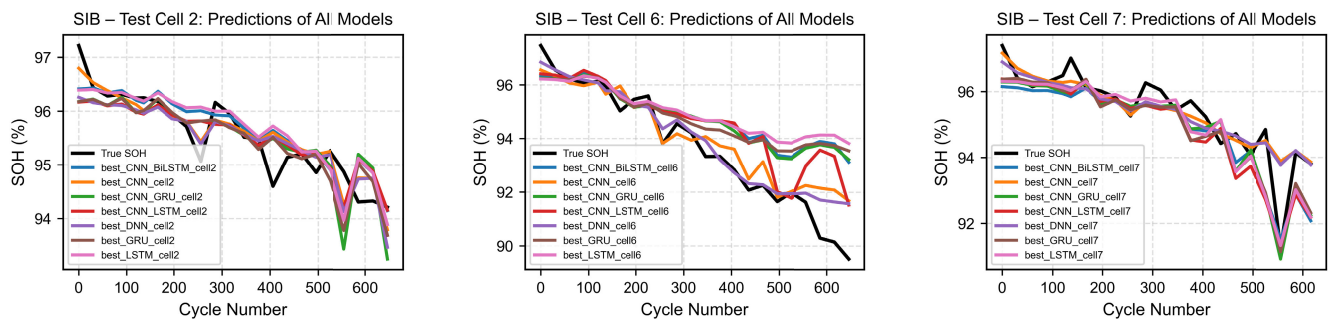


FIGURE 12. Predictions of all models for the three SIB test cells: Cell 2 (left), Cell 6 (center), and Cell 7 (right).

After identifying the best-performing LIB-only models (referred to as the *LIB baseline*), a transfer learning procedure was applied to evaluate the feasibility of cross-chemistry knowledge transfer from LIB to SIB. To ensure consistent input dimensionality across chemistries, the LIB ICA sequences were first padded to match the truncated length of the SIB datasets ($L_{SIB} = 3953$). The pretrained LIB

baseline models were loaded and retrained under an LOCO framework using a masking strategy. Their performance was benchmarked against the original unpadded LIB baselines to confirm that the padding–masking procedure did not degrade predictive accuracy. Only models exhibiting stable performance after padding were retained for the subsequent transfer learning stage.

In the transfer learning stage, all feature-extraction layers of the selected LIB models were frozen, and only the final fully connected layers (a 64-unit dense layer followed by a 1-unit output layer) were fine-tuned using SIB training samples, again under an LOCO framework. The resulting transfer-learned SIB models were finally evaluated against the corresponding SIB baseline models to quantify the effectiveness of transferring ICA-derived representations learned from LIBs to SIBs. Fig. 10 summarises the complete workflow adopted in this study.

IV. DISCUSSIONS AND RESULTS

Throughout this section, “LOCO-Cell X model” denotes the LOCO configuration in which Cell X is the held-out test cell and the remaining cells are used for training and validation.

A. HYPERPARAMETER OPTIMIZATIONS FOR LIB AND SIB

Tables 2 and 3 report the validation and LOCO test performance across LIBs (1, 3, 5, 8) and SIBs (2, 6, 7).

For LIBs, recurrent architectures (LSTM/GRU) and CNN-recurrent hybrid models consistently achieve superior performance compared to DNNs and standalone CNNs. For instance, the LSTM achieves a test MAE of 1.3267 ($R^2 = 0.8392$) for LOCO-Cell 1, while CNN-BiLSTM attains the best performance for LOCO-Cell 5 with a test MAE of 0.8811 ($R^2 = 0.9413$). However, performance degrades for more challenging cases such as Cell 8, where all models exhibit test MAE values above 2%. Overall, these results highlight the importance of temporal modeling in LIB SOH estimation.

In contrast, a different trend is observed for SIBs, where simpler architectures such as DNNs and CNNs consistently outperform recurrent and hybrid models. For example, CNN achieves a test MAE of 0.2591% ($R^2 = 0.8401$) for LOCO-Cell 2, while DNN performs best for LOCO-Cell 6 with a test MAE of 0.4281% ($R^2 = 0.9065$). Across all cases, recurrent-based models generally show higher prediction errors. This suggests that simpler models are sufficient to capture SIB degradation patterns without requiring complex temporal modeling.

This can be attributed to differences in ICA characteristics caused by different electrochemical characteristics. As discussed in Section II-B, LIBs exhibit sharp, well-defined peaks with smooth temporal evolution, leading to strong long-range dependencies that favor recurrent models. In contrast, SIB ICA profiles are broader, noisier, and exhibit less pronounced peak structures, making recurrent models more susceptible to overfitting. Consequently, simpler models such as CNNs and DNNs, which rely on static feature representations rather than modeling temporal dynamics, are less prone to overfitting and provide more robust performance for SIB.

These findings highlight the effectiveness of applying DL models for SOH estimation in both LIB and SIB systems. In this sense, such approaches can be integrated into BMS to enable accurate and reliable health assessment. Furthermore, the results emphasize the importance of model selection

Distribution of Δ MAE (Padding – Truncation) Across All Test Cells

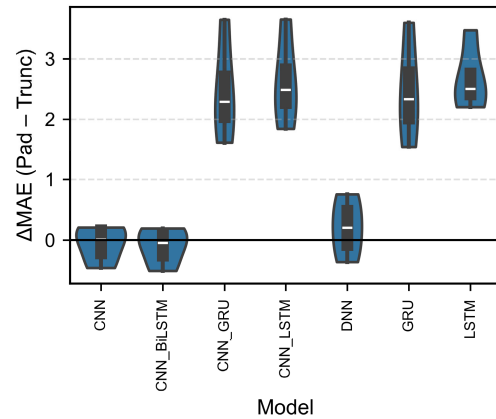


FIGURE 13. Distribution of the MAE difference between the padding method and the baseline method (padding minus baseline) across all test LIBs for each model.

based on the underlying electrochemical characteristics of the battery.

B. LOCO PREDICTION TRAJECTORIES FOR ALL MODELS

Fig. 11 shows the SOH prediction trajectories of all neural network models for four LIB test cells (1, 3, 5, and 8) under LOCO evaluation. In all cases, the predicted degradation curves closely match the actual SOH trends in the first half of the cycles. However, in the LOCO-Cell 1 and LOCO-Cell 8 models, the prediction accuracy decreases. Specifically, in the LOCO-Cell 1 model, Cell 8 is included in the training set, and Cell 8 exhibits the steepest SOH degradation among all cells; therefore, the model tends to overestimate the degradation in later ageing cycles. Conversely, in the LOCO-Cell 8 model, the model has never encountered such a large degradation curve during training, resulting in slightly higher predicted SOH values that gradually deviate from the actual SOH values in later cycles. Overall, these models are able to capture the global monotonic decrease and local fluctuations in SOH throughout the ageing process. Despite some minor biases within certain cycle ranges, the results demonstrate that ICA-derived features enable the models to reliably estimate SOH trajectories close to the true values.

Fig. 12 presents the SOH prediction results of all neural network models for the three SIB test cells (2, 6, and 7) under the LOCO evaluation. Similar to the observations for LIBs, the predicted trajectories closely follow the true SOH degradation patterns for most test cells. The only clear deviation occurs in the LOCO-Cell 6 model, where the models produce higher SOH values than the ground truth in the later ageing cycles. This behavior arises because Cells 2 and 7, which are used for training, exhibit slower degradation than Cell 6; consequently, the models tend to underestimate the degree of SOH decline when encountering the more rapidly degrading Cell 6. Overall, across the four LIB and three SIB test cells, the predicted SOH curves

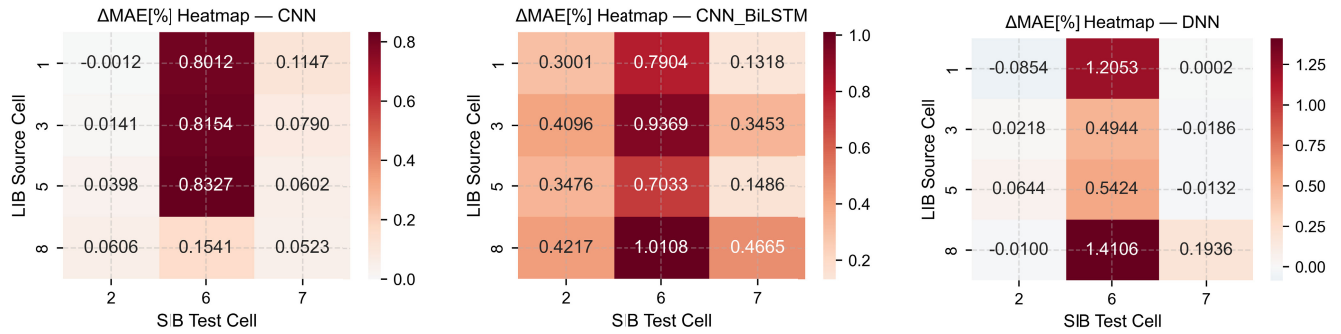


FIGURE 14. Transfer learning MAE difference heatmaps for CNN (left), CNN-BiLSTM (center), and DNN (right) models.

generally capture the global ageing trends and often reflect finer local variations as well. Although several models exhibit gradual deviations during the later stages of ageing, nearly all predictions remain closely aligned with the true SOH in the earlier cycles. These results confirm that the learned representations can generalize robustly to unseen SIBs, while also revealing that the models encounter limitations when the degradation behavior of the training cells differs substantially from that of the test cell.

C. PRIOR WORK BEFORE TRANSFER LEARNING

To identify the most robust LIB models for subsequent transfer learning to SIBs, the padding-and-retraining procedure described in Section III-C was first applied to all LIB baseline models.

Fig. 13 presents the distribution of ΔMAE (i.e., padded minus baseline performance) across all LIB architectures. A consistent pattern emerges: only three architectures exhibit stable behavior after padding: CNN, CNN-BiLSTM, and DNN. The accuracy of these models changes little, with many test units even showing slight improvements, indicating that their feature extraction mechanisms are largely insensitive to the newly added padding regions. In contrast, all recurrent-dominated architectures suffer from substantial performance degradation once padding is introduced. The hybrid CNN-GRU and CNN-LSTM models, as well as the purely recurrent GRU and LSTM models, display large positive ΔMAE values (typically between 2 and 3), confirming that recurrent sequence modelling is highly susceptible to long non-informative padded segments. This sensitivity results in a pronounced decline in prediction accuracy. Based on these observations, three representative architectures were selected for cross-chemistry transfer learning: CNN, CNN-BiLSTM, and DNN. To avoid instability effects interfering with the assessment of cross-chemical knowledge transfer, other models sensitive to padding were excluded.

D. TRANSFER LEARNING FROM LIB TO SIB

Figs. 14 show the ΔMAE heatmaps obtained when transferring models trained on LIBs (Cells 1, 3, 5, and 8) to SIBs

(Cells 2, 6, and 7). The metric in each grid is defined as

$$\Delta\text{MAE} = \text{MAE}_{\text{TL}} - \text{MAE}_{\text{SIB-baseline}}, \quad (5)$$

where MAE_{TL} is the real error after transfer learning and $\text{MAE}_{\text{SIB-baseline}}$ is the real error of a native SIB-only model trained using the same LOCO protocol. Negative values indicate performance improvement due to transfer learning, while positive values correspond to accuracy degradation.

Across all three models (CNN, DNN, and CNN-BiLSTM), a consistent cross-chemistry transfer pattern is observed under the LOCO-Cell configurations. LOCO-Cell 6 (SIB) consistently yields the largest degradation across all tested models, with $\Delta\text{MAE} = 0.154\% - -1.411\%$, consistent with its highest baseline MAE (Section IV-B), indicating a strong distribution mismatch between the ICA characteristics of SIB Cell 6 and those of the LIBs. In contrast, near-zero or slightly negative ΔMAE values are observed for transfers from LOCO-Cell 1 and 3 (LIB) to LOCO-Cell 2 (SIB), indicating effective cross-domain generalization. Similarly, LOCO-Cell 7 (SIB) remains stable for most models, with intermediate difficulty observed only for the CNN-BiLSTM architecture (0.467% for transfer from LOCO-Cell 8).

Among the three models, CNNs exhibit the most stable cross-domain behavior across all LOCO LIB-to-SIB combinations, with relatively low ΔMAE variation. Notably, CNN is the only architecture that achieves $\Delta\text{MAE} < 1\%$ for the challenging LOCO-Cell 6 (SIB) transfer. DNNs are the most sensitive to source-target mismatch: they perform well for LOCO-Cell 2 and LOCO-Cell 7 (SIB), while the largest degradation ($\Delta\text{MAE} = 1.411\%$) is observed for LOCO-Cell 6 (SIB). In contrast, the CNN-BiLSTM architecture shows moderate performance across all combinations, with minimum ΔMAE values remaining above 1.3%.

Overall, these results show that (i) the feasibility of LIB-to-SIB transfer learning depends strongly on the target SIB cell in the LOCO-Cell evaluation (ii) CNN-based feature extractors provide superior robustness to cross-chemistry domain shift and (iii) LIB and SIB ICA pattern share partially transferable electrochemical structures, especially for SIBs whose degradation trajectories are geometrically aligned with those of the LIB source cells. In this sense, LIB-to-SIB

transfer learning not only provides a practical solution to data scarcity in emerging SIB systems, but also serves as a tool to uncover cross-chemistry electrochemical similarities between LIB and SIB.

V. CONCLUSION

This work presented a comprehensive degradation campaign covering both LIB and SIB, including characteristic Testing (1C), HPPC test and ageing procedure, yielding a unified dataset, from which SOH trajectories and ICA representations were systematically extracted. A unified deep-learning framework was established for SOH estimation, supported by chemistry-specific hyperparameter optimization under an LOCO evaluation protocol. The results show that ICA curves provide a reliable representation of degradation dynamics, enabling accurate SOH prediction and offering BMS a robust solution for real-time monitoring and decision support. The framework can be trained offline and deployed for real-time inference. In particular, lightweight architectures such as CNNs and DNNs are well suited for TinyML applications, enabling efficient deployment on resource-constrained embedded hardware through model quantization and further optimization. Such characteristics are highly desirable in BMS, where continuous real-time monitoring, fast response, and low-power operation are essential for ensuring safe and reliable system performance. Moreover, prediction accuracy also varies across LOCO test cases. When the test cell exhibits behavior that deviates substantially from the training and validation cells, all models experience a degradation in performance, underscoring the need to expand the diversity and representativeness of cell datasets to mitigate generalization limitations.

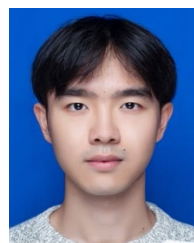
In the transfer-learning analysis, only the CNN, DNN, and CNN-BiLSTM models trained on LIB data demonstrated stable behavior under the padding/masking adjustment required for SIB ICA sequences. Moreover, TF from LIB models and fine-tuning only the final dense layers on SIB data proved feasible, reflecting the partially aligned electrochemical characteristics between LIBs and SIBs. This further reveals the potential to leverage the abundant LIB data to support SIB applications. Nevertheless, several inconsistent results were observed for specific target cells, indicating that transfer learning may be limited when substantial discrepancies exist between the degradation patterns of the source and target cells. Interestingly, this limitation closely aligns with the behavior observed in the SIB baseline models, where performance degradation similarly arose whenever the training and test cells exhibited substantial differences. This consistency shows that the present study is based on a limited number of cells (four LIB and three SIB), and thus the findings on model performance and transferability are preliminary and may be influenced by the specific degradation patterns of the limited cell set. Future work will involve extending the degradation campaign to a broader range of cells and ageing conditions, allowing the development of more

generalizable SOH estimators and a deeper understanding of cross-chemistry transferability.

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